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Discovering Invariant Patterns of Cognitive Decline Via an Automated Analysis of the Cookie Thief Picture Description Task

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Abstract

The Cookie Thief task has been extensively adopted to uncover patterns characterizing Alzheimer's Disease (AD). Yet, how findings reported on this task generalize across different corpora remains unclear. The main objective of this study was to assess the cross-corpora robustness of 145 cognitive, linguistic, and acoustic features, conducting a statistical and correlation analysis with clinical scores. Thus, four corpora were considered. The experimental results indicate that cognitive features reflecting attention focus and linguistic features representing information units and vocabulary richness displayed a robust behavior across corpora and reported a strong correlation with clinical scores. However, paralinguistic features such as pause length and pitch variability reported a lack of homogeneous behavior. This study provides insights into inter-corpora dissimilarities and highlights which features can be used when applying models trained in one domain to an unseen cohort or domain.

1. Introduction

Dementia affects over 6 million elderly individuals in the United States [1]. The global annual medical cost of Alzheimer's disease (AD) is currently around 1 trillion, which is expected to almost double by 2030. Currently, dementia diagnosis relies on clinical examination, cognitive testing, labs, and imaging studies, all of which are typically time-consuming and costly. Innovative screening tools for early detection of dementia using reliable biomarkers are becoming critical to intervening in the early stages of the disease [2, 3]. Individuals with dementia commonly experience language disorders [4], exhibiting differences in language-based cognitive tasks, including word-search ability, language fluency, and sentence repetition, compared to healthy controls (CTLs) [5]. Therefore, speech-based analysis can be a useful tool to detect dementia in its early stages. Over the past three decades, researchers have investigated speech-based dementia detection using the Cookie Theft Picture (CTP) description task [6]. In this task, participants are presented with the CTP from the Boston Aphasia test (see Fig. 1) and are asked to verbally describe everything they observe in the picture [7]. Various studies have introduced methods for dementia and AD detection using speech and language-based features, yielding promising results [8, 9, 10, 11]. Nevertheless,

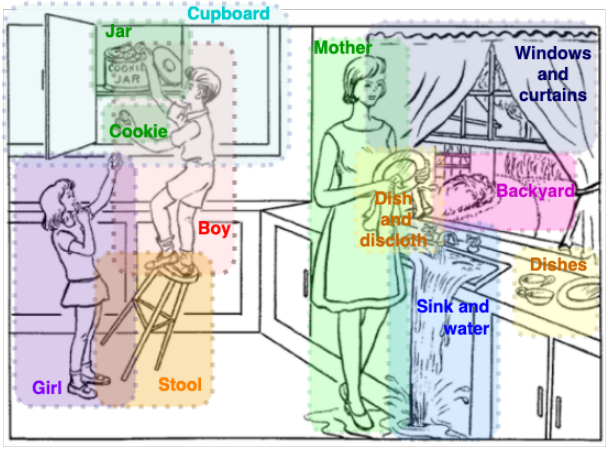


Figure 1: The Cookie Theft Picture from the Boston Diagnostic Aphasia Examination. The main Areas of Interest (AoIs) are emphasized with colored rectangles.

the corpora adopted in these studies are often very limited in size and representative population, which may limit the generalizability of the results. Arguably, limitation of the generalizability of findings and methods is one of the critical issues to solve before implementing speech and machine learning based solutions in clinical applications.

Very few studies conduct cross-corpora comparisons to examine interpretable features that may characterize AD. A recent cross-corpora study by Linsday et al. [12] analyzed features for language impairment in AD from 154 speakers (78 healthy subjects, 76 patients with AD) of two different languages (106 English-speaking and 47 French-speaking individuals). The study confirmed AD patients' semantic impairment in terms of low information efficiency and semantically empty language. Ablimit et al. [13] presented a comprehensive analysis of speech and language features derived from two distinct corpora: the Interdisciplinary Longitudinal Study on Adult Development and Aging (ILSE) [14], a German data set comprising over 6500 hours of biographic interviews from 1000 participants recorded over two decades, and a cross-sectional data set

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from the ADReSS challenge 2020 [15]. While the classification experiments showed promising results, the variability in the best classifiers and feature sets across corpora raises questions about the transferability of methods across different languages, tasks, and recording conditions. Other studies investigated features' robustness by mainly undertaking cross-corpora classification experiments, considering two or three corpora at a time and experimenting with various interpretable and non-interpretable representations, respectively [16, 17, 18]. However, as computational techniques and machine learning methods advance, the complexity of understanding classifiers' decisions and the associated clinical assumptions increases. Therefore, good machine learning performance alone does not necessarily entail clinical evidence for language impairment as a cognitive symptom in AD and dementia, in general. Additional methodological precautions must be taken to ensure that the results are clinically valid, generalizable, and do not overfit to a single corpus or language.

In general, it is unclear how previous findings generalize across corpora, as variability can be introduced by different background noises, sampling frequency, microphones, and task protocols that can influence the estimation of the various measurements adopted [19, 20]. The few studies assessing the robustness of the interpretable features of AD exhibit some limitations. First, most studies mainly conducted cross-corpora classification experiments but did not analyze individual features' robustness, which is necessary to identify which features have an invariant behavior across corpora. Second, their analysis mainly considered a limited number of corpora and features at a time, which limited the power of the study. Third, the criteria adopted to probe the robustness of the features were too weak, increasing the probability of false reporting as robust features that display different behaviors across cohorts and corpora. To overcome these limitations, in this study, the behaviors of 145 interpretable features encoding cognitive, linguistic, and acoustic information were examined using four corpora recorded under various acoustic conditions and protocols. The proposed analysis leverages *robustness* [21, 20] as a criterion in which a feature behaves the same, independently of the corpus considered. Identifying which features are robust represents a step forward in the development of a *universal* framework for the evaluation and monitoring of AD. Furthermore, given their interpretable meaning, if these features display robustness, they might be employed in clinical scenarios as AD biomarkers.[1]

2. Materials

Four different corpora were considered: NeuroLogical Signals (NLS) [22, 21, 23], ADReSS-M [24], Delaware Corpus [25] and English Lu Corpus [25]. Except for NLS, all corpora are from the Dementia Bank and are available at `https://dementia.talkbank.org`.[2] Table 1 summarizes the demographics of the corpora and statistics of the severity of the disease.

2.1. NeuroLogical Signals

NeuroLogical Signals (NLS) is a data set collected at Johns Hopkins Medicine (JHM) by the authors of this study. It contains recordings from individuals with different neurological disorders and CTLs, all English native speakers. The Johns Hopkins Medical Institutional Review Board approved the data collection, and all participants signed informed consent. Participants were recorded with a headset at 24 kHz in a quiet room. In this study, we considered 14 participants with mild cognitive impairment (MCI) due to AD (biomarker confirmed), 6 with AD, and 33 CTLS of the same age. None of the participants in the CTL group had a history of symptoms related to AD or any other neurological disorder.

2.2. ADReSS-M

ADReSS-M (Multilingual Alzheimer's Dementia Recognition through Spontaneous Speech) is an ICASSP 2023 Signal Processing Grand Challenge that aims to explore the extraction of universal acoustic features from speech data to facilitate multilingual detection of AD [24]. The ADReSS-M corpus consists of audio recordings of picture descriptions obtained from 148 AD patients and 143 CTLs, in English and Greek languages. The corpus is divided into three splits: an English training split, a Greek sample split, and a Greek test split. In this study, only the training split was considered, as the test set does contain speech samples elicited from the CTP task. The English training set was collected from 122 AD patients and 115 CTLs. Participants were asked to describe the CTP in English during the recording session. Audio recordings are distributed in MPEG audio layer 2/3 format, with a sample rate of 44 kHz.

2.3. The Delaware Corpus

The Delaware corpus in the DementiaBank [25] contains recordings collected over the Internet during 2021–2022 through Zoom from 46 individuals with MCI and 24 age- and education-matched CTLs, all English native speakers. Language samples were obtained using the DementiaBank protocol, available from `https://dementia.talkbank.org/protocol/`. Audio recordings are distributed in MPEG audio layer 2/3 format, with a sample rate of 32 kHz. In this study, only speech samples elicited from the CTP task were considered.

2.4. English Lu Corpus

The English Lu corpus consists of recordings of the CTP description task from 16 CTLs and 26 individuals with either dementia, AD, MCI, or vascular dementia.[3] It is a browsable corpus with downloadable transcripts and videos. Audio recordings are distributed in MPEG audio layer 2/3 format, with a sample rate of 48 kHz.

3. Methods

3.1. Data Pre-processing

All recordings were resampled at 16 kHz as required by the algorithms/models used for feature extraction. Although transcriptions and word alignments are available in some of the original corpora (i.e., Delaware and Lu corpus), transcriptions and word alignments were generated using WhisperX [26] for all corpora considered [4] This decision was made to ensure fair

[1] The code to reproduce our experiments can be found at the following link: `https://github.com/Annafavaro/Alzheimers_Features`.

[2] DementiaBank was supported by NIH grants AG005133 and AG003705 to the University of Pittsburgh.

[3] The data is in audio format and has a DOI: doi:10.21415/4KE0-6348.

[4] The large-v2 checkpoint available on the Hugging Face Hub was used. See `https://huggingface.co/openai/whisper-large-v2`

Corpus	Category	Sample size (n)	Age (std)	Female (%)	MoCA (std)	MMSE (std)
NLS	AD + MCI	20	70.70 (8.20)	25	20.63 (6.19)	–
	CTL	33	68.27 (6.54)	67	27.12 (1.323	–
ADReSS-M	AD	122	69.37 (6.85)	65	–	17.84 (5.46)
	CTL	115	66.06 (6.28)	65	–	29.02 (1.14)
Lu Corpus	AD+MCI+VD	28	76.32 (17.00)	54	–	–
	CTL	26	76.46 (18.32)	69	–	–
Delaware	MCI	51	73.45 (7.57)	57	24.10 (2.47)	–
	CTL	31	68.51 (5.77)	83	27.54 (1.54)	–

Table 1: Demographics and disease severity profile of the study cohorts. Statistics include sample size, age distribution, gender breakdown (as a percentage of females), scores from the Mini-Mental Status Examination (MMSE), and the Montreal Cognitive Assessment (MoCA), when available. Abbreviations: VD: Vascular Dementia; MCI: Mild Cognitive Impairment.

Feature family	Description
Acoustic (16)	1-8: Prosodic features (e.g., pitch variability) 9-16: Articulatory features (e.g., speech rate)
Cognitive (28)	17-19: Time spent on AoIs (tot, avg, std) 20-22: # of visits to AoIs (tot, avg, std) 23-25: Time spent on animate AoIs (tot, avg, std) 26-28: # of visits to animate AOIs (tot, avg, std) 29-31: Time spent on inanimate internal AoIs (tot, avg, std) 32-34: # of visits to inanimate internal AoIs (tot, avg, std) 35-37: Time spent on inanimate external AoIs (tot, avg, std) 38-40: # of visits to external internal AoIs (tot, avg, std) 41: Time to approach the first AoI 42: Time elapsed from last to first AoI 43-44: Transition time (avg, std)
Linguistic (101)	45-84: Information units (e.g., % of IUs mentioned) 85-90: Surface features (e.g., # words) 91-108: syntactic complexity indices (e.g., # coordinate phrases per T-unit) 109-134: part-of-speech based (e.g., # of verbs) 135-139: Lexical Richness (e.g., corrected type-token ratio) 140-145: Age of acquisition features (e.g., Total Kuperman's AoA)

Table 2: Summary of the interpretable features extracted, subdivided by feature family. In parentheses, the total number of coefficients extracted for each family. Abbreviations: tot, total; avg, average; std, standard deviation; POS: Part of Speech; AoA: Age of Acquisition; AoI: Area of Interest; IU: Information Unit.

comparability of feature analysis across the various corpora. Moreover, the Root Mean Square (RMS) loudness normalization procedure was applied before extracting intensity-related features (see section 3.2).

3.2. Features

Table 2 summarizes the set of features considered in the analysis. All the features were extracted from speech samples elicited during the CTP description task. We limited our analysis to features that have a clear interpretable meaning and that can be used in clinical scenarios as biomarkers of AD.

3.2.1. Acoustic Features

To characterize the effect of AD on prosody, we analyzed features based on pitch and intensity, such as the standard deviation of the fundamental frequency and the standard deviation of intensity. These features were calculated using Parselmouth [5], a Python library for the Praat software.[6] The autocorrelation method was used to perform pitch analysis. We also considered prosodic features related to pause, such as speech time, total pause time, percentage pause time, mean pause duration, silence-to-speech ratio, and pause variability. To perform the extraction, we used DigiPsychProsody [7], a Python library that computes the features mentioned above using the WebRTC Voice Activity Detector [8]. 20 ms was used as frame length for voice activity detection. A total of 8 prosodic features were extracted.

With respect to articulation, we analyzed features representing the speech rate, the average syllable duration, and the articulation rate. The speech rate was calculated as the number of syllables divided by the total duration of the audio, including pauses. The articulation rate was calculated as the number of syllables divided by the total speech time (pause excluded). The average duration of the syllable was calculated as the total speaking time divided by the number of voiced peaks. We also calculated the *voiced fraction* representing the ratio of voiced frames to the total number of frames in the pitch analysis. Finally, the stability of vocal tract control was measured with the standard deviation of the first four formants (i.e., F1, F2, F3, and F4). All these features were extracted using Parselmouth[9] The formants of each recording were extracted with a maximum formant value of 5.5 kHz, a window length of 25 ms, a time step of 6.25 ms, and a pre-emphasis of 50 kHz. A total of 8 articulatory features were extracted.

3.2.2. Alignment-based Features

Along with language, visuospatial function is affected early in AD. This is manifested in the form of degraded attentional and visual search, among other disturbances [27]. Previous studies analyzing eye movement behavior during static image search reported that people with dementia fixated on fewer Areas of Interest (AoIs) for a significantly shorter duration compared to CTLs and that the time to first fixation within the AoIs was significantly longer in people with AD than in CTLs

[5] https://parselmouth.readthedocs.io/en/stable

[6] https://www.fon.hum.uva.nl/praat/

[7] https://github.com/NeuroLexDiagnostics/DigiPsych_Prosody

[8] https://github.com/wiseman/py-webrtcvad

[9] In particular, the following libraries were used: https://github.com/uzaymacar/simple-speech-features and https://github.com/drfeinberg/PraatScripts/tree/master.

Info Unit	Synonyms
Animate AoIs	
Boy	boy, son, brother, male child
Girl	girl, daughter, sister, female child
Woman	mom, mother, lady, parent, female, adult, grownup
Inanimate internal AoIs	
Kitchen	kitchen, room
Cookie	cookie, biscuit, cake, treat
Jar	jar, container, crock, pot
Stool	stool, seat, chair, ladder
Sink	sink, basin, washbasin, washbowl, washstand, tap
Plate	plate
Dishcloth	dishcloth, dishrag, rag, cloth, napkin, towel
Water	water, dishwater, liquid
Window	window, frame, glass
Cupboard	cupboard, closet, shelf
Dishes	dish, dishes, cup, cups, counter
Curtains	curtain, curtains, drape, drapes, drapery, blind, blinds, screen, screens
Inanimate external AoIs	
Exterior	exterior, outside, garden, yard, outdoors, backyard, driveway, path, tree, bush
Actions	
Steal	take, steal, taking, stealing
Fall	fall, falling, slip, slipping
Wash	wash, dry, clean, washing, drying, cleaning
Overflow	overflow, spill, overflowing, spilling

Table 3: A list of the information units and their synonyms.

[28, 29, 30, 31]. Inspired by previous studies on eye-movement and speech analysis [32], we designed a set of word alignment-based features to estimate individuals' focus of attention while describing the CTP. To do so, we first identified the AoIs in the image using the list of *information units* proposed by Croisile et al. [33]. We considered AoIs all the nouns in the list denoting animate and inanimate entities. WordNet [34] was used to create a set of synonyms, hypernyms, and hyponyms for each AoI. For the complete list of AoIs considered, see Table 3. Using WhisperX automatic speech recognizer, we generated the word time stamps (both start and end times) for each word in an utterance. Transcriptions and word timestamps were used to extract the following features: time spent in the AoIs (estimated using the length of the uttered word), time to approach the first AoI (estimated using word start time), number of visits to the AoIs (i.e., number of times a given AoI is mentioned), the transition time from one AoI to another, and time elapsed between the first and last AoI mentioned. Features were computed considering all the animate AoIs, all the inanimate AoIs internal to the house, all the inanimate AoIs external to the house, and all the AoIs collectively (see Table 3). We considered AoIs of different natures separately, as previous studies reported that subjects with AD tended to spend less time than CTLs on specific AoIs (e.g., windows, curtains) [35, 36]. Except for the features representing the time to approach the first AoI and the time elapsed between the last and first AoI, we computed the sum, mean, and standard deviation for each feature. Thus, a total of 28 features were considered.

3.2.3. *Information Units Features*

Considering the list of Croisile et al. [33], information units can be nouns (e.g., jar, cookie, boy, kitchen) or verbs (e.g., taking, drying). As in [37], for each information unit, we extracted two features: a binary feature indicating whether the subject has mentioned that specific information unit (or any of its synonyms) and a frequency count of how many times an information unit has been mentioned. Moreover, features representing the total number of information units, the ratio between keywords related to information units and words unrelated to information units, and the percentage of total information units (including synonyms) mentioned in a given narrative were considered. The complete list of information units considered and their synonyms is shown in Table 3. A total of 40 features were extracted.

3.2.4. *Surface Features*

The surface feature set contains features that do not belong to any specific linguistic branch (e.g., lexico-semantics, syntax). They are features such as the total number of characters, the total number of words, and the total number of sentences. A total of 6 surface features were extracted. All the surface features were extracted using the Linguistic Feature ToolKit (LFTK) [38].

3.2.5. *Syntactic Features*

To represent syntactic complexity, we extracted measures of the complexity of an utterance such as the depth of the parse tree, the mean length of T-unit[10], the mean length of clause, and the number of clauses per sentence. To extract these features, we used NeoSCA, a fork of the L2 Syntactic Complexity Analyzer[11]. A total of 18 syntactic complexity descriptors were extracted. We also considered part-of-speech based features that count part-of-speech properties on the token level based on dependency parsing. These features represent the total and unique number of various part-of-speech in a given narrative. Some examples are the total and unique number of nouns, auxiliaries, proper nouns, conjunctions, and verbs. We used spaCy's dependency parser for English [12] to extract these features. A total of 26 part-of-speech-based features were considered.

3.2.6. *Lexical-semantics Features*

Five metrics were calculated to capture the richness of vocabulary in a text. Besides the standard type-token ratio metric, we used normalized metrics less sensitive to the length of the text, such as root type-token ratio, corrected type-token ratio, logarithmic type-token ratio, and uber type-token ratio [40]. Moreover, we also computed two metrics related to the age of acquisition of words [41]: Kuperman's age of acquisition of words and Brysbaert's age of acquisition of words [42, 43]. They measure how old people are on average when they first learn a given word. Another feature is the word frequency statistics based on SUBLTEXus, a database that compiles word frequencies from subtitles of movies and TV series in American English, augmented with the Zipf scale, a standardized frequency measure characterized by a logarithmic scale ranging approximately from 1 to 6 or 7 [44]. Age-of-acquisition-based features were computed considering the whole narrative and per word. A total of 6 Age-of-Acquisition-related features were extracted. All lexical-semantic features were extracted using LFTK [38].

[10] A T-unit, as defined by Hunt [39], consists of a dominant clause accompanied by any dependent clauses linked to it.

[11] https://github.com/tanloong/neosca

[12] urlhttps://spacy.io/models/en

3.3. Statistical and Correlation Analysis

The non-parametric pair-wise Mann-Whitney U tests [45] of significance were performed to determine whether there were significant differences between the AD and the CTL group for each feature. To perform the pair-wise Mann-Whitney U tests, we used *scipy.stats.mannwhitneyu* library in Python. To control the False Discovery Rate, we applied the Benjamini–Hochberg correction[13]. As family-wise error rate, α was set to 0.05. Moreover, the correlations of the feature values with the Montreal Cognitive Assessment (MoCA) scores and Mini-Mental State Examination (MMSE) scores (when available) were assessed using the *scipy.stats.spearmanr* library in Python. Before the correlation analysis, features were normalized by subtracting the mean and dividing by the standard deviation.

3.4. Analysis of Robustness

We considered *robust* a feature if these three conditions are satisfied:

1. If it reports a statistically significant difference between the medians of the AD and CTL group distributions in at least *two* corpora.
2. If it displays the same expected behavior (EB) in *all* the corpora where it is significant.
3. If the correlation between the feature values and the clinical scores (when available) is significant, the correlation trend follows the EB postulated for the feature.

Similar robustness criteria were previously introduced by Favaro et al. [21, 20]. The EB of a given feature is grounded in the previous literature documenting the speech dysfunctions connected to AD. An ideal robust feature is a feature that exhibits homogeneous behavior *across corpora*.

4. Results and Discussion

Table 4 summarizes the pairwise Kruskal–Wallis $H-$test results for those features that were statistically significant (p <0.05) and displayed a robust behavior across corpora. For some of these features, the distributions are displayed in the boxplots reported in Figure 2. With respect to the family of cognitive features, subjects with AD tended to focus for a significantly shorter amount of time (tot, avg) on salient AoIs than CTLs, in particular to those located externally to the house, and to visit less animate and inanimate external AoIs than CTLs. These behaviors were observed on the NLS, ADReSS-M, and Lu corpus. Moreover, compared to the CTL group, the AD group reported a greater variability in the transition time from one AoI to another on the NLS and ADReSS-M corpus. Results from the statistical analysis were corroborated by those reported within the correlation analysis. Namely, on the ADReSS-M corpus, the features representing the time spent on AoIs (tot, avg) and the number of visits to AoIs (tot, avg) were positively correlated with the MMSE scores. Overall, these results are consistent with previous studies showing that search patterns of people with AD are significantly less focused and organized than those of CTLs and that they tend to look at significantly fewer AoIs for a significantly shorter duration than CTLs [28, 31, 36].

For the family of linguistic features, the syntactic features representing the number of clauses, dependent clauses, and complex nominals were significantly lower for the AD group compared to the CTL group on the NLS and Lu corpus. Similarly, syntactic features based on part-of-speech, such as unique nouns, number of nous, and determiners, were significantly lower for the AD group vs. the CTL group on all corpora considered, except on the Delaware corpus. With respect to information-units-related features, metrics such as the count of general information units and the percentage of information units mentioned were significantly lower for the AD group than for the CTL group on the NLS and ADReSS-M corpus. These two metrics also reported a significant positive correlation with the MMSE scores on the ADReSS-M corpus (as expected). Interestingly, aligned with the findings of [35] et al., subjects with AD tended to mention the curtains and inanimate objects located outside the house less than CTLs. Moreover, on both NLS and the Lu Corpus, the total Kuperman's age of acquisition and the total Brysbaert's age of acquisition were significantly lower for ADs than for CTLs. This result suggests that people affected

[13]To perform the Benjamini–Hochberg correction, we used *statsmodels.stats.multitest.fdrcorrection*, with default method.

	NLS		ADReSS-M		Lu Corpus	
Feature name (EB)	***p*-value (OB)**	**AUROC**	***p*-value (OB)**	**AUROC**	***p*-value (OB)**	**AUROC**
Cognitive						
Tot. time spent in AoIs (↓)	ns (↓)	0.54	0.001 (↓)	0.62	0.01 (↓)	0.73
Avg. time spent in AoIs (↓)	ns (↓)	0.53	<0.001 (↓)	0.66	0.01 (↓)	0.71
Tot. # of visits to AoIs (↓)	0.03 (↓)	0.68	<0.001 (↓)	0.70	<0.001 (↓)	0.81
Tot. time spent in AoI external AoIs (↓)	0.03 (↓)	0.67	<0.001 (↓)	0.66	0.02 (↓)	0.66
Tot. # of visits to external AoIs (↓)	0.03 (↓)	0.67	<0.001 (↓)	0.66	0.02 (↓)	0.66
Std. transition time (↑)	0.049 (↑)	0.65	<0.001 (↑)	0.66	ns (↓)	0.55
Linguistic						
# of clauses (↓)	0.008 (↓)	0.71	0.001 (↓)	0.51	0.03 (↓)	0.71
# of dependent clauses (↓)	0.02 (↓)	0.69	ns (same)	0.51	0.03 (↓)	0.71
# of complex nominals (↓)	0.004 (↓)	0.73	ns (↓)	0.51	0.005 (↓)	0.78
# of unique nouns (↓)	0.001 (↓)	0.77	<0.001 (↓)	0.73	0.003 (↓)	0.80
# of nouns (↓)	0.002 (↓)	0.76	<0.001 (↓)	0.62	0.002 (↓)	0.82
# of determiners (↓)	0.01 (↓)	0.71	<0.001 (↓)	0.66	0.002 (↓)	0.83
Tot. count IUs (↓)	0.02 (↓)	0.68	<0.001 (↓)	0.71	ns (↓)	0.82
% of IUs (↓)	0.03 (↓)	0.67	<0.001 (↓)	0.74	0.001 (↓)	0.82
Tot. Kuperman AoA (↓)	0.002 (↓)	0.74	ns (↓)	0.56	0.007 (↓)	0.76
Tot. Brysbaert AoA (↓)	0.002 (↓)	0.76	ns (↓)	0.54	0.01 (↓)	0.75
Tot. Subtlex Zipf (↓)	0.003 (↓)	0.75	ns (↓)	0.54	0.009 (↓)	0.76
CTTR (↓)	ns (↓)	0.64	0.001 (↓)	0.71	0.03 (↓)	0.71
RTTR (↓)	ns (↓)	0.64	0.001 (↓)	0.71	0.03 (↓)	0.71
Acoustic						
Voiced fraction (↓)	<0.001 (↓)	0.74	0.002 (↑)	0.63	ns (↑)	0.53

Table 4: Summary of the main significant features from the statistical analysis. The corpus name, feature name, p-value, expected behavior (EB), observed behavior (OB), and AUROC are reported for each significant feature. Abbreviations: ns: not significant; AoIs: Areas of Interest; IUs: Information Units; AoA: Age of Acquisition; CTTR: Corrected Type-Token Ratio; RTTR: Root Type-Token Ratio.

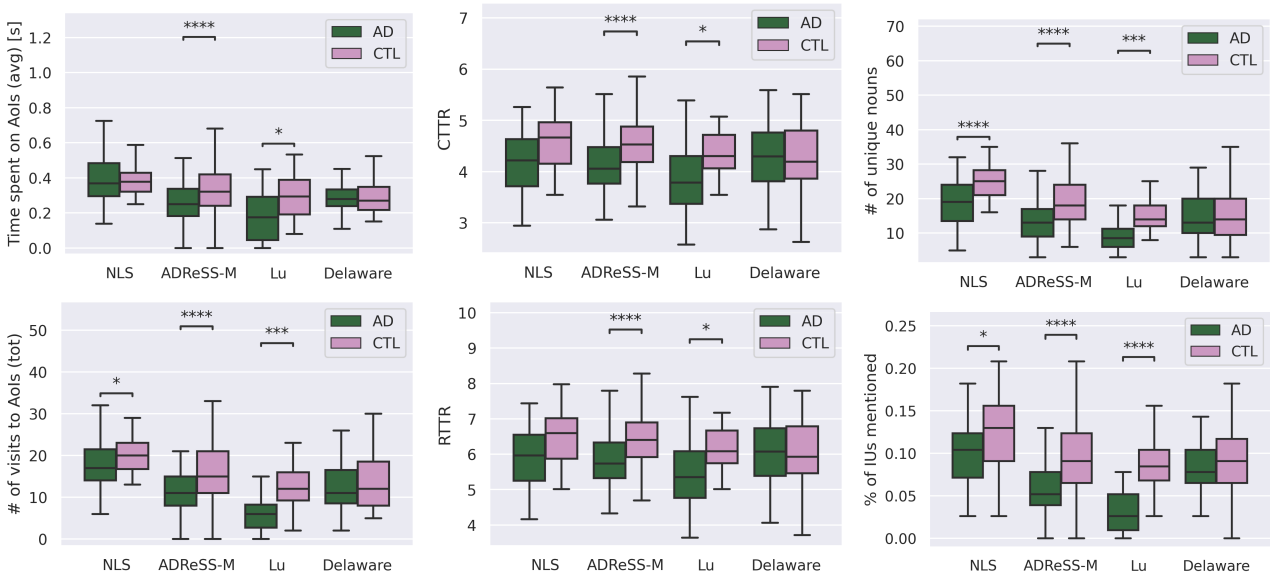


Figure 2: Boxplots of some of the significant features from the statistical analysis. Asterisks highlight statistically significant differences in the median between groups, where * means $0.01 < p \leqslant 0.05$, ** means $0.001 < p \leqslant 0.01$, *** means $0.0001 < p \leqslant 0.001$, and **** means $p \leqslant 0.0001$. Abbreviations: AoIs: Areas of Interest; CTTR: Corrected Type-Token Ratio; RTTR: Root Type-Token Ratio; IUs: Information Units.

by dementia tend to adopt simpler words acquired earlier in life than CTLs. Similarly, lexical richness measures, such as corrected type-token ratio and root type-token ratio, were significantly lower for people with AD than for CTLs on the Lu and ADReSS-M corpus. These two metrics also reported a significant positive correlation with the MMSE scores on ADReSS-M, indicating that vocabulary size is negatively affected by the progression of dementia.

For the family of acoustic features, subjects with AD tended to report a significantly lower fraction of voiced segments than CTLs in the NLS and ADReSS-M corpus. However, most acoustic features did not display robust behaviors across corpora. For example, while the standard deviation of the pitch was significantly lower for subjects with AD on NLS, the same feature was significantly higher for the AD group on the ADReSS-M corpus. Moreover, while the speech rate was significantly lower on the NLS corpus for the AD group compared to the CTL group, it was significantly higher on the ADReSS-M and Lu corpus. Furthermore, as expected, features related to pause time, pause duration, and pause variability negatively correlated with MoCA scores on NLS. However, these features showed significantly higher values on the NLS and ADReSS-M corpus and significantly lower values for the same group on the Lu Corpus. This behavior on the Lu-corpus may stem from the abundance of background chatter and music present in recordings from the AD group, which, in turn, could have adversely impacted the accuracy of the voice activity detection system utilized for feature extraction. The only acoustic feature displaying an invariant cross-corpora behavior was *voiced fraction*.

In general, the inconsistencies observed in the various acoustic features may stem from variations in the acoustic environments during data collection. For example, the Delaware corpus was collected during Zoom sessions, which can take place in diverse settings, often with little control over ambient noise, room acoustics, and other environmental factors. The contradictory outcomes reported in the acoustic analysis align with previous findings. Ablimit et al. [13] previously showed that the feature representing the silence rate displayed opposite distributions on the ADReSS and ILSE corpus, respectively. Similarly, Lindsay et al. reported a lack of cross-corpora robustness of the acoustic features in the characterization of AD. Other previous studies analyzing the cross-corpora robustness of acoustic features in characterizing Parkinson's disease showed how variations in recording conditions introduce potential sources of error or differences in the obtained values when assessing vocal characteristics [46, 19, 20], which makes it important to consider and account for specific recording conditions when interpreting acoustic measurements. To conclude, it is worth mentioning that none of the features analyzed were significant on the Delaware corpus. One plausible explanation for this phenomenon, among others, could be that subjects in this corpus have Mild Cognitive Impairment (MCI) and thus exhibit very mild signs of dementia. To the best of our knowledge, this is the first study to analyze the behavior of various features in this recently published corpus.

5. Conclusions

This study examined a composite set of interpretable features encoding cognitive, acoustic, and linguistic information. It aimed to examine the effectiveness of these features in modeling characteristic patterns of dementia by assessing their discriminatory power and cross-corpora robustness. Our analysis showed that cognitive features reflecting the number of visits to the areas of interest, the time spent on the areas of interest, and linguistic features related to information units and vocabulary richness were robust across corpora and strongly correlated with clinical scores. However, almost none of the acoustic features passed the test of robustness. These results are consistent with previous studies showing a lack of robustness in the paralinguistic descriptors of AD, which emphasizes the importance of considering recording conditions when analyzing acoustic features. In future work, we plan to further validate our results by analyzing a larger sample of cohorts and exploring corpora in different languages. We will also conduct cross-corpora machine learning experiments following a rigorous methodology [47] to observe to which extent the lack of robustness reported in this study affects the model performances.

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